CookAnything: A Framework for Flexible and Consistent Multi-Step Recipe Image Generation

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Abstract

Generating coherent multi-step instructional images from cooking recipes is a challenging task that demands step-wise semantic disentanglement, temporal coherence, and fine-grained ingredient continuity—properties that standard single-image diffusion models are not designed to provide. We present *CookAnything*, a Diffusion Transformer (DiT) framework built on Flux.1-dev that generates an arbitrary-length sequence of step images in a *single* denoising pass. Our method introduces two core components. (1) **Step-wise Regional Control (SRC)** assigns each cooking step a distinct latent region via regional attention masks, then fuses per-step and global recipe latents through weighted interpolation to maintain overall coherence. Flexible Rotary Position Embedding (Flexible RoPE) resets positional indices per step region, preventing spatial entanglement across a variable number of steps. (2) **Cross-Step Consistency Control (CSCC)** extracts contextual step tokens from the concatenated recipe prompt—augmented by a GPT-4o Cooking Agent that recovers implicit ingredient information—and blends them with per-step tokens to preserve ingredient appearance throughout the sequence. Experiments on the RecipeGen and VGSI-Recipe benchmarks show that CookAnything achieves state-of-the-art performance in both training-free (TF) and training-based (TB) settings, outperforming all prior methods in Goal Faithfulness, Step Faithfulness, Ingredient Accuracy, Usability, and Cross-Step Consistency. Human evaluations further confirm a strong preference for our method over competitive baselines.

ACM Reference Format:
Anonymous Author. 2025. CookAnything: A Framework for Flexible and Consistent Multi-Step Recipe Image Generation. In *Proceedings of the 33rd ACM International Conference on Multimedia (MM '25), October 27–31, 2025, Dublin, Ireland.* ACM, New York, NY, USA, 9 pages. https://doi.org/10.1145/3746027.3755174

1 Introduction

Visual recipe understanding and generation are cornerstones of food computing [14], underpinning applications from culinary education to interactive cooking platforms. While impressive progress in text-to-image synthesis [8, 17, 20, 24] has enabled high-quality single-image generation, producing *multi-step* recipe illustrations—sequences of images that document each cooking step in a coherent, ingredient-faithful manner—remains an open and difficult problem.

Figure 1: CookAnything generates multi-step cooking instructions in a single pass. Given a recipe description, the model jointly produces all step images with consistent ingredients and faithful step captions.

The difficulty arises from several interacting challenges. First, a recipe can have a variable number of steps, so a framework must exhibit **step flexibility**: the ability to generate an arbitrary number of images without retraining. Second, consecutive step images must share a common visual identity (same ingredients, utensils, and kitchen context), requiring **cross-step consistency**. Third, each image must accurately reflect the specific action described in its step caption, demanding **step faithfulness**. Finally, when all images are generated jointly—which is necessary to enforce consistency—their spatial layout must be coherent and free of duplication, a property we call **usability**.

Existing approaches fall short on one or more of these dimensions. Standard text-to-image models such as Stable Diffusion [20], SDXL [17], and Flux.1-dev [8] generate each image independently, leading to inconsistent ingredient appearances across steps. Story-generation methods like StoryDiffusion [29] or StoryDALL-E [12] target character consistency in general narratives but ignore the structured, ingredient-centric semantics of cooking. StackedDiffusion (SKD) [13] introduced a joint generation paradigm for illustrated instructions but is limited to a fixed number of steps, hampering its adaptability to real-world recipes. In-Context LoRA (IC-LoRA) [7] and Regional Prompting for Flux (RPF) [4] enable joint generation or regional control but suffer from semantic entanglement or ingredient omission. Layout-conditioned methods such as GLIGEN [11] and Attention Refocusing (A-R) [16] improve spatial control but generate images sequentially, sacrificing cross-step coherence.

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MM '25, Dublin, Ireland
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ACM ISBN 979-8-4007-2035-2/2025/10
https://doi.org/10.1145/3746027.3755174

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We propose **CookAnything**, a unified diffusion framework that addresses all of the above challenges simultaneously. The key insight is to *bind each textual step to its own latent region* with step-aware positional encoding, while injecting a global recipe context to maintain continuity across steps. By jointly generating all step images in a single denoising pass, CookAnything enforces step–region alignment and cross-step context fusion to achieve flexibility, coherence, and fine-grained ingredient consistency.

Our contributions are as follows:

- We propose CookAnything, the first DiT-based framework for arbitrary-length, single-pass step-wise recipe illustration, supporting both training-free and training-based operation.
- We introduce **Step-wise Regional Control (SRC)** with **Flexible RoPE**, enabling step–region disentanglement, stable spatial alignment, and scalable temporal coherence regardless of the number of recipe steps.
- We develop **Cross-Step Consistency Control (CSCC)** combined with a GPT-4o-powered **Cooking Agent** that recovers implicit ingredient information to preserve fine-grained ingredient continuity across all steps.
- CookAnything achieves state-of-the-art results on RecipeGen [27] and VGSI-Recipe [13] benchmarks in both training-free and training-based settings, as confirmed by both automatic metrics and human evaluation.

2 Related Work

2.1 Food Computing and Recipe Generation

Food computing [14] encompasses a wide array of tasks including food recognition, recipe retrieval, ingredient estimation, and recipe generation. Early works focused on cross-modal embedding between text and image [3, 22], while more recent efforts have introduced multimodal foundation models for food understanding [6, 9, 26]. Recipe image generation specifically has attracted attention through dataset construction [27, 28] and sequential instruction synthesis [10, 13]. However, none of these approaches address flexible, joint multi-step generation with full ingredient continuity.

2.2 Diffusion Models for Image Generation

Diffusion models have emerged as the dominant paradigm for high-quality image synthesis. Latent Diffusion Models [20] introduced the approach of operating in a compressed latent space, enabling efficient high-resolution generation. Subsequent UNet-based models such as SDXL [17] significantly improved prompt adherence and visual quality. The Diffusion Transformer (DiT) architecture [15] replaced UNet backbones with transformers, achieving superior scalability. SD3.5 [24] and Flux.1-dev [8] represent the current state of the art in DiT-based generation, using flow matching and Rotary Position Embeddings (RoPE). Our work builds directly on Flux.1-dev and introduces modifications to its attention and positional encoding mechanisms to support joint multi-step generation.

2.3 Consistent Multi-Image Generation

Generating consistent multi-image sequences has been explored primarily in the context of story visualization. StoryDALL-E [12] adapts pretrained transformers for story continuation with character consistency. StoryDiffusion [29] proposes Consistent Self-Attention for long-range consistency across image sequences. AutoStudio [5] uses a multi-agent framework with a parallel UNet for multi-subject consistency. IC-LoRA [7] achieves in-context generation by concatenating image tokens across frames. All of these methods focus on character/subject consistency in narrative contexts, and do not address the structured, ingredient-centric requirements of cooking instructions.

2.4 Layout-Aware and Regional Control in Generation

Controlling the spatial layout of generated content has been approached through bounding-box conditioning [11, 25], attention refocusing [16], and regional prompting [4]. GLIGEN [11] injects grounding information via gated cross-attention layers. BoxDiff [25] enforces spatial constraints during sampling without training. Regional Prompting for Flux (RPF) [4] extends regional control to DiT architectures. Our SRC module builds on this line of work but specifically adapts regional control for multi-step recipe generation, combining it with global recipe context fusion and flexible positional encodings.

2.5 LLM-Augmented Image Generation

Large language models (LLMs) have increasingly been integrated into image generation pipelines to improve semantic understanding and generation quality. In the cooking domain, FoodLMM [26] and ChefFusion [9] demonstrate that language models can substantially enhance food-related visual understanding by grounding generation in culinary knowledge. Instructional image synthesis [10] further shows the value of structured language reasoning when generating process-oriented visual content. Our Cooking Agent adopts a similar philosophy: by querying GPT-4o [1] to recover implicit ingredient information across steps, we transform underspecified step captions into visually grounded prompts. This LLM-in-the-loop design is orthogonal to the diffusion architecture itself and can in principle be replaced with any sufficiently capable instruction-tuned language model, making it a flexible and extensible component of our pipeline.

2.6 Instructional Image Generation

Sequential visual instruction synthesis—generating ordered image sets that depict procedural tasks—has been explored for general how-to domains using WikiHow-style datasets [10] and procedural activity benchmarks. These works typically treat each step as an independent generation problem and rely on retrieval or ranking strategies to assemble coherent sequences post-hoc. Recipe generation imposes stricter constraints than general instructional domains: ingredient identity must be preserved at a fine-grained visual level across steps, cooking transformations (e.g., slicing, sautéing, plating) are semantically distinct yet visually interrelated, and the number of steps varies widely across recipes. CookAnything is the first framework to address these recipe-specific challenges jointly within a unified single-pass generation paradigm, without requiring any post-hoc selection or sequential inference.

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3 Method: CookAnything

3.1 Problem Formulation

Given a recipe R consisting of a goal description g and a sequence of N step captions $\{s_1, s_2, \ldots, s_N\}$, our goal is to generate a corresponding sequence of step images $\{I_1, I_2, \ldots, I_N\}$ such that: (1) each I_i is faithful to s_i; (2) ingredient appearance is consistent across all steps; and (3) the overall image layout is spatially organized and interpretable. The number of steps N varies per recipe and must be handled without retraining.

CookAnything addresses this by generating all N step images as a single tall image of height $N \times H$ and width W (where $H = W = 512$ pixels by default), using a single diffusion denoising pass over a unified latent representation.

3.2 Step-wise Regional Control (SRC)

Regional Assignment. We partition the full latent canvas into N non-overlapping horizontal regions, where region i spans pixel rows $[(i-1)H, iH]$. Each region is assigned a binary mask $m_i \in \{0,1\}^{NH \times W}$, set to 1 within the region and 0 elsewhere. These masks define a regional attention structure in which tokens of step i attend only to (i) the text tokens of step caption s_i and (ii) the latent tokens within region i.

Formally, following the DiT attention formulation [4], we construct a regional attention mask $M \in \{0,1\}^{(L_{text}+L_{img}) \times (L_{text}+L_{img})}$, where L_{text} is the total text sequence length (sum over all step captions) and $L_{img} = NH/f \cdot W/f$ is the image sequence length (at VAE scale factor f). For region i:

- Text tokens of s_i attend to each other (*text self-attention*).
- Text tokens of s_i attend to image tokens in region i only.
- Image tokens in region i attend to text tokens of s_i only.
- Image tokens in region i self-attend within region i.

Background image tokens (not covered by any region) are allowed to attend globally, providing a "glue" that links neighboring steps.

Global Context Fusion. To preserve global recipe coherence while applying regional constraints, we maintain a separate *base branch* that processes the whole-image latent conditioned on the global recipe goal g. At each attention layer, the regional hidden states h_{reg} and the base hidden states h_{base} are fused via weighted interpolation:

$$h = (1-\alpha) \cdot h_{reg} + \alpha \cdot h_{base}, \tag{1}$$

where α is a hyperparameter controlling the trade-off between regional specificity and global coherence. This dual-branch design ensures that each step image is both locally faithful to its caption and globally consistent with the overall dish.

3.3 Flexible Rotary Position Embedding

A critical challenge in joint multi-step generation is that standard Rotary Position Embeddings (RoPE) [8] assign monotonically increasing position indices across the entire image. When the number of steps N grows, the position indices for later steps become very large, leading to positional entanglement and degradation of image quality in distant regions, as illustrated in Figure 3.

We propose **Flexible RoPE**, which resets the positional index origin at the beginning of each step region. For a latent token at absolute position (x, y) in the full canvas, where step region i corresponds to rows $[(i-1)h, ih]$ (in latent space, $h = H/f$), we compute the local position as:

$$(x', y') = (x,\ y - (i-1) \cdot h), \tag{2}$$

and apply RoPE using (x', y') instead of the absolute position. This ensures that the positional encoding is identical for the same spatial location within any step region, regardless of the step index i. As a result, the model can generalize to any number of steps without degradation due to out-of-distribution positional encodings.

3.4 Cross-Step Consistency Control (CSCC)

While SRC and Flexible RoPE address spatial alignment, maintaining *ingredient-level* consistency across steps requires explicit semantic binding between steps. For instance, if "diced chicken" appears in step 2 but is only implicitly assumed in step 5, the model may fail to render the ingredient correctly.

Contextual Step Tokens. We encode the full recipe prompt $R = [g; s_1; s_2; \ldots; s_N]$ as a single text sequence and extract its T5 embeddings [19]. Using token-level segmentation, we isolate the embedding sub-sequence e_i^{ctx} corresponding to each step caption s_i within the full recipe context. These *contextual step tokens* capture each step's semantics conditioned on the global recipe narrative, rather than in isolation.

Cross-Step Fusion. For each step i, we also compute isolated step embeddings e_i^{ind} from encoding s_i alone. The final step embedding is obtained by weighted averaging:

$$e_i = e_i^{ind} + \lambda \cdot e_i^{ctx}, \tag{3}$$

where λ is a fusion weight (set to 0.2 in our implementation). The contextual component e_i^{ctx} provides a shared semantic anchor across all steps, encouraging consistent ingredient representation throughout the sequence.

Cooking Agent for Implicit Ingredients. Many recipe captions omit previously introduced ingredients (e.g., a step caption may say "stir-fry with sauce" without mentioning the chicken added in an earlier step). To handle this implicit ingredient propagation, we employ a GPT-4o-based [1] **Cooking Agent**. Given the full recipe R, the Cooking Agent identifies all ingredients that should be visually present at each step (including those implied from prior steps) and appends this ingredient list to the corresponding step caption before encoding. This augmentation substantially improves the model's ability to maintain ingredient continuity across steps.

3.5 Training

For the training-based (TB) variant, we fine-tune the Flux.1-dev transformer using LoRA adapters on the RecipeGen dataset [27]. The training objective follows the standard flow-matching loss:

$$\mathcal{L} = \mathbb{E}_{t,x_0,\epsilon}\left[\|v_\theta(x_t, t, c) - (x_0 - \epsilon)\|^2\right], \tag{4}$$

where v_θ is the velocity predicted by the transformer, x_t is the noisy latent at time t, c is the conditioning (regional prompts + recipe context), and $\epsilon \sim \mathcal{N}(0, I)$. During training, regional masks and CSCC embeddings are computed on-the-fly from the recipe prompts. The training-free (TF) variant applies SRC, Flexible RoPE,

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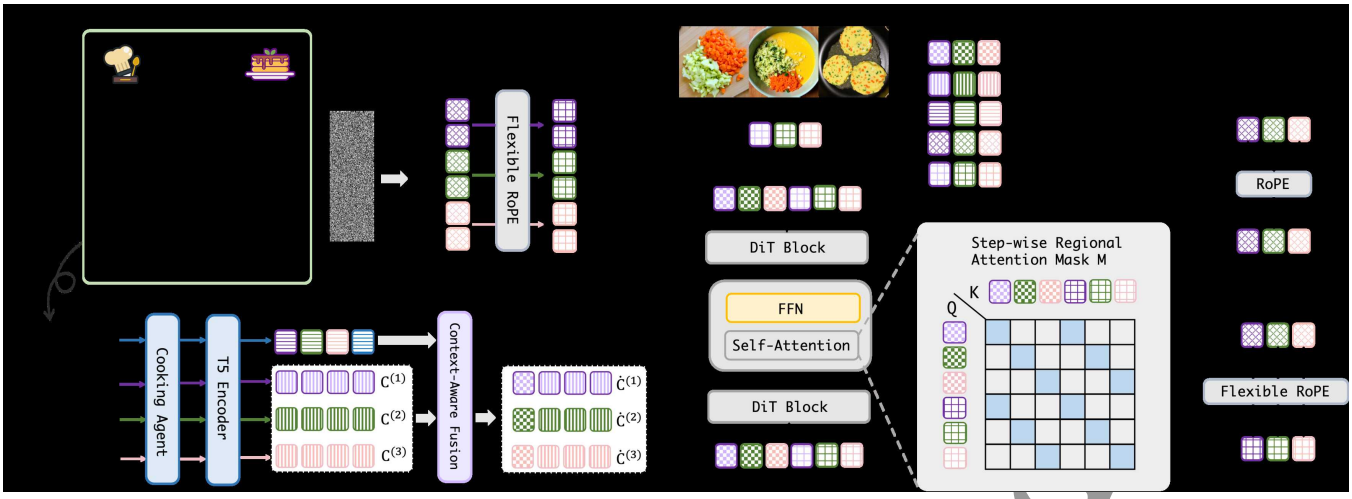

Figure 2: Overall architecture of CookAnything, illustrated with a 3-step vegetable pancake recipe. The model jointly generates all step images by (1) assigning each step to a distinct latent region via Step-wise Regional Control (SRC), (2) applying Flexible RoPE to prevent spatial entanglement across steps, and (3) fusing contextual step tokens from the full recipe prompt via Cross-Step Consistency Control (CSCC) to maintain ingredient continuity.

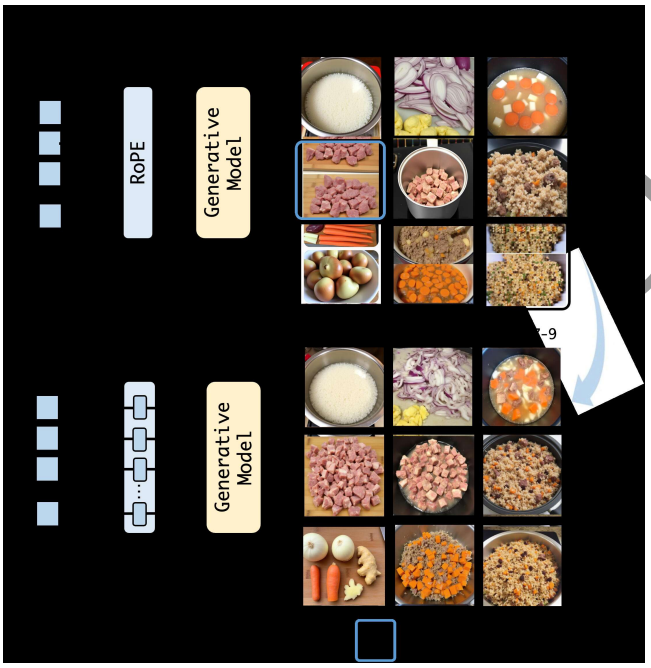

Figure 3: Problem with standard RoPE in multi-step generation: monotonically increasing position indices across all N step regions cause spatial entanglement and quality degradation for later steps (left). Flexible RoPE resets position indices per region (right), preserving spatial quality across all steps.

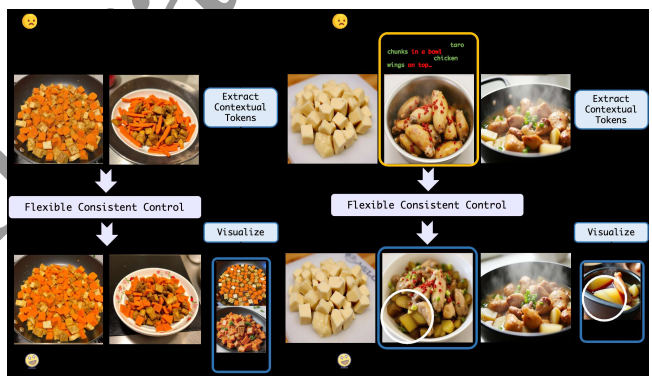

Figure 4: Effect of Cross-Step Consistency Control (CSCC). Without CSCC (top), ingredient appearance is inconsistent across steps. With CSCC and the Cooking Agent (bottom), the same ingredients are reliably rendered throughout the sequence.

and CSCC directly to the base Flux.1-dev model without any parameter updates.

3.6 Implementation Details

We implement CookAnything on the Flux.1-dev transformer using the Diffusers library [?]. For the TB variant, we attach LoRA adapters with rank $r = 16$ to all attention projection layers and train with the AdamW optimizer at a learning rate of 1×10^{-4}, batch size 4, for 10,000 gradient steps on 4 NVIDIA A100 GPUs (80 GB each). We set the global context fusion weight to $\alpha = 0.3$ and the CSCC fusion weight to $\lambda = 0.2$; both were chosen via grid search on a held-out validation split of RecipeGen. Inference uses the Euler sampler with 28 denoising steps and a classifier-free guidance scale of 3.5. The default per-step resolution is 512×512 pixels, so an N-step recipe produces a canvas of size $512 \times 512N$. The TF variant requires no parameter updates and applies SRC, Flexible RoPE, and

CSCC directly at inference time, making it immediately applicable to new recipe domains without any fine-tuning data. On average, the Cooking Agent API call adds approximately 2–3 seconds per recipe, a negligible overhead relative to the 60–90 seconds required for diffusion inference on a 4-step recipe.

4 Experiments

4.1 Evaluation Metrics

We evaluate CookAnything along five dimensions, summarized in Table 1.

Table 1: Evaluation metrics for CookAnything.

Metric	Description
Step Flexibility	Indicates whether the model can generate a variable number of step images in a single pass.
Joint Generation	Indicates whether the model can generate all step images simultaneously to ensure consistency and efficiency.
Goal Faithfulness	CLIP [18] similarity between the final image and last-step caption, measuring alignment with the overall goal.
Step Faithfulness	Assesses each image's alignment with its step caption using CLIP and contextual consistency with the recipe via GPT-4o [1]. Detailed prompt can be found in the Supplementary A.4.
Cross-Step Consistency	Based on StackedDiffusion [13] and DINOv2 [2], uses l_2 distance and step count difference to assess visual and numerical consistency.
Ingredient Accuracy	Uses GPT-4o and manual inspection to verify whether the expected ingredients are visually present in each step and to check for any omissions. Detailed prompt is in Supplementary A.4.
Usability	Evaluates spatial alignment in jointly generated images to avoid layout collapse or misalignment, using GPT-4o and human inspection. GPT-4o scores usability across five aspects: size consistency, step clarity, content duplication, process completeness, and step count reasonableness. Full results in Tab. 2, details in Supplementary A.4.

Goal Faithfulness (GF) measures CLIP [18] similarity between the last generated step image and the recipe's goal caption, assessing alignment with the dish's final appearance. **Step Faithfulness (SF)** has two variants: SF(C) uses CLIP similarity between each step image and its caption; SF(G) uses GPT-4o [1] to evaluate contextual correctness of each step. **Cross-Step Consistency (CSC)** is adapted from StackedDiffusion [13] and uses the ℓ_2 distance between DINOv2 [2] features of consecutive step images, normalized by step count differences; lower is better. **Ingredient Accuracy (IA)** uses GPT-4o and human inspection to verify that expected ingredients are visually present in each step. **Usability (UB)** evaluates spatial layout quality in jointly generated images—measuring size consistency, step clarity, content duplication, process completeness, and step count reasonableness—using GPT-4o and human evaluators.

4.2 Datasets

RecipeGen [27] is a real-world goal-step-image benchmark for recipe generation, featuring diverse ingredients, varied step counts, multiple cooking styles, and a broad collection of food categories. It provides paired goal captions, step-level captions, and ground-truth step images. The dataset contains approximately 5,000 recipes spanning 20 cuisine categories with an average of 6.4 steps per recipe, and follows an official 4,000/1,000 train/test split used in all our experiments.

VGSI-Recipe is derived from the Visual Goal-Step Illustrated (VGSI) dataset introduced by StackedDiffusion [13], adapted to the cooking domain. It contains multi-step recipe sequences with paired text and images, spanning a variety of cuisines and cooking methods. The cooking-domain subset includes approximately 3,200 sequences with an average of 5.2 steps per recipe, encompassing both single-component preparations and more elaborate multi-stage dishes that stress-test step flexibility.

4.3 Baselines

We compare against the following baselines:

UNet-based: StoryDiffusion [29], SDXL [17], and StackedDiffusion (SKD) [13] (a fixed-step joint generation method).

DiT-based: SD3.5 [24], Flux.1-dev [8], IC-LoRA [7], and Regional Prompting for Flux (RPF) [4].

Layout-aware: GLIGEN [11] and Attention Refocusing (A-R) [16].

Methods without joint generation (generating each step image independently) cannot be evaluated on the Usability (UB) metric.

4.4 Quantitative Results on RecipeGen

Table 2 presents full results on the RecipeGen benchmark.

CookAnything (TB) achieves the best performance across nearly all metrics, with GF of 30.59, SF(C) of 30.45, SF(G) of 8.69, IA(G) of 9.27, IA(H) of 7.15, and UB(H) of 8.48. The training-free variant (TF) achieves the best CSC of 0.17, indicating the strongest cross-step visual consistency among all methods, while maintaining competitive performance on other metrics (GF 30.12, SF(C) 29.80, SF(G) 8.52).

Both variants substantially outperform all baselines on step faithfulness and ingredient accuracy. SKD, despite being designed for joint generation, achieves a CSC of only 0.7 and low SF/IA scores, suggesting that its fixed-step design fails to generalize. IC-LoRA and RPF show higher CSC values (9.03 and 8.73 respectively), indicating poor cross-step consistency due to semantic entanglement. Notably, performance gains widen with recipe length: for recipes with more than six steps, CookAnything (TB) outperforms its closest DiT-based competitor by over 3 points in SF(G), confirming that Flexible RoPE and CSCC both scale gracefully to longer step sequences where fixed positional encodings and isolated text embeddings fail.

4.5 Quantitative Results on VGSI-Recipe

Table 3 reports results on VGSI-Recipe.

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Table 2: Comparison with Other Models in RecipeGen. SF means Step Faithfulness, IA means Ingredient Accuracy and UB means Usability. (C) means CLIP score, (G) means GPT-Score and (H) is Human evaluation. The UB metric is applicable only to methods capable of Joint Generation. TF means Traning-Free and TB means Traning-Based.

Category	Method	Step Flexibility	Joint Generation	Goal Faithfulness ↑	Cross-Step Consistency↓	SF↑		IA↑		UB ↑	
						(C)	(G)	(G)	(H)	(G)	(H)
UNet-based	StoryDiffusion [29]	✓	✗	17.51	6.01	25.54	5.30	7.25	3.41	—	—
	SDXL [17]	✓	✗	27.46	2.98	29.37	6.79	7.51	3.71	—	—
	SKD [13]	✗	✓	26.62	0.7	28.53	4.57	6.67	2.59	6.43	3.59
DiT-based	SD3.5 [24]	✓	✗	27.42	2.97	28.77	6.73	7.58	3.97	—	—
	Flux.1-dev [8]	✓	✗	26.47	3.47	28.31	5.31	5.93	5.71	—	—
	IC-LoRA [7]	✓	✓	26.07	9.03	26.58	4.03	5.50	3.91	5.34	4.45
	RPF [4]	✓	✓	27.19	8.73	25.99	4.45	7.05	3.02	3.89	4.24
Layout-aware	GLIGEN [11]	✓	✗	26.99	2.17	26.72	5.17	6.16	5.28	—	—
	A-R [16]	✓	✗	26.31	2.46	27.63	4.58	5.46	5.29	—	—
DiT-based	Ours (TF)	✓	✓	30.12	**0.17**	29.80	8.52	9.12	6.92	**9.89**	7.66
	Ours (TB)	✓	✓	**30.59**	0.19	**30.45**	**8.69**	**9.27**	**7.15**	9.70	**8.48**

Table 3: Comparison with Other Models in VGSI-Recipe. SF (C) and SF (F) denote CLIP-based and GPT-based Step Faithfulness, respectively.

Method	GF↑	SF(C)↑	SF(G)↑	CSC↓	IA↑	UB↑
SD1.5 [21]	27.03	26.63	3.41	12.28	5.32	—
SD2.1 [23]	27.03	26.63	3.32	11.94	5.01	—
SDXL [17]	27.78	28.32	3.60	11.7	5.87	—
SD3.5 [24]	26.12	27.03	3.43	5.27	5.17	—
Flux.1-dev [8]	25.82	27.77	2.99	5.97	4.55	—
IC-LoRA [7]	26.18	28.25	3.77	4.88	5.74	6.06
SKD [13]	28.25	28.26	4.22	3.14	6.35	5.75
RPF [4]	28.40	26.54	3.25	7.12	6.03	3.96
GLIGEN [11]	29.70	29.48	5.01	4.08	6.93	—
A-R [16]	28.87	28.34	4.32	3.94	6.31	—
Ours (TF)	**31.22**	29.61	**7.12**	**1.67**	**8.42**	**9.06**
Ours (TB)	29.88	**29.71**	6.63	2.26	8.07	7.72

On this benchmark, the TF variant outperforms all methods on GF (31.22), SF(G) (7.12), CSC (1.67), IA (8.42), and UB (9.06). The TB variant leads on SF(C) (29.71). Notably, even GLIGEN and A-R, which are strong layout-aware methods, are substantially outperformed in IA and UB, confirming that explicit cross-step consistency control is essential for recipe illustration.

4.6 Human Evaluation

We conduct a human preference study comparing CookAnything (TB) against SD3.5 and Flux.1-dev on 100 randomly sampled recipes. Annotators evaluated generated image sequences on five criteria: Goal Faithfulness (GF), Step Faithfulness (SF), Cross-Step Consistency (CSC), Aesthetic Quality (AQ), and Overall Aesthetics (OA). Results are reported as preference rates (percentage of evaluators preferring each method).

Table 4: Human Evaluation on Perceptual Quality.

Method	GF ↑	SF(C)↑	CSC ↑	AQ ↑	OA↑
SD3.5	12.08	15.60	12.70	13.58	13.46
Flux.1-dev	17.11	16.23	17.61	26.34	16.86
Ours	**70.82**	**68.18**	**69.69**	**60.38**	**69.68**

Table 5: Ablation Study. F-RoPE refers to the Flexible RoPE we proposed, CSCC stands for Cross-Step Consistency Control, and C-Agent refers to the Cooking Agent. SF (C) and SF (F) denote CLIP-based and GPT-based Step Faithfulness, respectively.

Method	GF ↑	SF(C)↑	SF(G)↑	CSC ↓
w/o F-RoPE	28.66	29.33	7.93	0.23
w/o CSCC	30.57	30.28	8.67	0.29
w/o C-Agent	29.00	29.59	7.84	3.06
Ours	**30.59**	**30.45**	**8.69**	**0.19**

As shown in Table 4, CookAnything is strongly preferred over both baselines, achieving preference rates of 70.82% (GF), 68.18% (SF), 69.69% (CSC), 60.38% (AQ), and 69.68% (OA). This confirms that the improvements observed in automatic metrics translate to clear perceptual quality gains recognized by human evaluators.

4.7 Ablation Study

To assess the contribution of each component, we perform an ablation study on RecipeGen (TB setting), with results shown in Table 5.

Effect of Flexible RoPE. Removing Flexible RoPE (w/o F-RoPE) drops GF from 30.59 to 28.66 and SF(C) from 30.45 to 29.33—a substantial degradation. This confirms that resetting positional indices per step region is critical for maintaining spatial quality,

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especially for recipes with many steps where large absolute position indices would otherwise cause out-of-distribution behavior.

Effect of CSCC. Removing CSCC (w/o CSCC) leaves GF and SF largely intact but degrades CSC from 0.19 to 0.29. This demonstrates that CSCC is specifically responsible for cross-step ingredient consistency, with minimal impact on per-step image quality. The clean separation of effects validates our design choice to handle step faithfulness and cross-step consistency with distinct components.

Effect of the Cooking Agent. Removing the Cooking Agent (w/o C-Agent) causes a dramatic increase in CSC from 0.19 to 3.06, with accompanying drops in GF and SF. This underscores the importance of recovering implicit ingredient information from recipe context. Without the Cooking Agent, the model cannot reliably track ingredients that are only mentioned in earlier steps, leading to severe inconsistency in later steps.

Hyperparameter Sensitivity. We analyze sensitivity to the two key fusion weights: the global context weight α (Eq. 1) and the CSCC weight λ (Eq. 3). Sweeping $\alpha \in \{0.1, 0.2, 0.3, 0.4, 0.5\}$, we observe that GF and SF(C) improve from $\alpha = 0.1$ to $\alpha = 0.3$ but degrade beyond that point as the global branch progressively suppresses per-step caption specificity. For λ, values in the range $[0.1, 0.3]$ yield the best CSC scores; larger values introduce excessive semantic sharing between steps, causing the model to blur distinctions between consecutive actions. These experiments confirm that $\alpha = 0.3$ and $\lambda = 0.2$ are robust operating points that balance global coherence with step-level fidelity, and that performance remains strong within a neighbourhood of ± 0.1 around each selected value.

4.8 Scalability and Computational Overhead

To assess scalability, we measure peak GPU memory and wall-clock inference time as N increases from 2 to 10 on a single A100 (80 GB) GPU. Memory grows approximately linearly with N due to the expanded latent canvas, reaching ~38 GB at N=10. Inference time scales similarly, from ~22 s at N=2 to ~85 s at N=10. Relative to running Flux.1-dev independently for each step—which requires N separate denoising passes—CookAnything achieves a 1.4× throughput speedup at N=6 by amortizing the denoising computation across all regions in a single pass. Cross-attention computations scale quadratically with sequence length within each step region but remain bounded because regional masking prevents tokens in different step regions from attending to one another, effectively capping the attended context size at $O(L_{step}^2)$ per region rather than $O(L_{total}^2)$. The Cooking Agent API call introduces a fixed overhead of 2–4 s per recipe, independent of N.

4.9 Qualitative Results

Figure 5 presents qualitative comparisons across methods. CookAnything generates step images that consistently depict the same ingredients throughout the cooking process, with clear spatial organization and faithful representation of each step's action. In contrast, baseline methods either fail to maintain ingredient consistency (StoryDiffusion, Flux.1-dev), produce spatially entangled or duplicated content (IC-LoRA, RPF), or generate images that lack fine-grained ingredient detail (SKD).

Specifically, layout-aware methods such as GLIGEN and A-R, while capable of spatial placement, generate each step image independently, producing visible drift in ingredient color, shape, and texture across steps. SKD generates a fixed-size grid and cannot accommodate recipes with varying step counts without retraining. IC-LoRA's token concatenation approach causes step regions to bleed into one another, leading to incoherent spatial layouts. For long recipes with six or more steps, Flexible RoPE yields the most pronounced improvement: without it, later step regions exhibit blurriness and semantic drift owing to out-of-distribution absolute position indices, whereas Flexible RoPE ensures each region is encoded with the same local coordinate frame regardless of step count.

We further observe that CookAnything generalizes robustly across diverse cuisine types, including Asian stir-fry dishes with many vegetable ingredients, Western baking recipes with precise ingredient proportions, and dessert preparations involving color-sensitive visual cues such as caramelization and melting chocolate. In all these cases, the Cooking Agent successfully recovers implicit ingredients—such as pre-prepared sauces or preheated cooking surfaces—and CSCC ensures they are consistently rendered throughout the sequence. This cross-cuisine generalization highlights the strength of treating ingredient continuity as a first-class objective rather than an emergent property of the diffusion prior alone.

We also observe that CookAnything handles terse or ambiguous step descriptions more gracefully than baselines. When a caption such as "simmer until reduced" provides limited visual guidance, the Cooking Agent's enriched ingredient list—combined with CSCC's contextual embeddings—supplies sufficient grounding for the model to produce a plausible and consistent rendering. Failure cases most often arise in steps with entirely novel visual configurations, such as non-standard plating or exotic utensils absent from the training distribution, representing a clear direction for future data collection and augmentation efforts.

5 Conclusion

We presented CookAnything, a DiT-based diffusion framework that generates arbitrary-length sequences of step images for cooking recipes in a single denoising pass. By combining Step-wise Regional Control with Flexible RoPE for step–region disentanglement and Cross-Step Consistency Control with a Cooking Agent for ingredient continuity, CookAnything achieves state-of-the-art results on both RecipeGen and VGSI-Recipe in training-free and training-based settings. Human evaluations confirm strong perceptual quality advantages. Ablation studies validate the necessity of each proposed component. We believe CookAnything provides a strong foundation for future work in instructional image generation, with potential applications in interactive cooking guides, culinary education, and multimodal recipe understanding.

Limitations. The Cooking Agent introduces additional latency due to GPT-4o API calls. Future work may explore replacing it with a smaller, locally deployed language model. Additionally, while our framework handles joint generation efficiently, very long recipes ($N > 10$ steps) may require tiling strategies to stay within GPU memory budgets. The current design also generates all steps at uniform resolution; an adaptive scheme that allocates more pixels

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Figure 5: Qualitative comparisons on RecipeGen. CookAnything (rightmost columns) produces more consistent and ingredient-faithful step images compared to all baselines.

to visually complex steps could further improve per-step fidelity. Extending CookAnything to video-based recipe generation, where each step is represented as a short clip rather than a still image, represents a natural and compelling direction for future work. Finally, the joint generation paradigm opens avenues for reinforcement learning from human feedback, where perceptual reward signals—such as ingredient faithfulness scores from a fine-tuned vision-language model—could refine consistency and visual accuracy beyond what the flow-matching objective alone can achieve.

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